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Inventor(s)	Khawas; Arnab et al.

Low area and power multi-bit flip-flop

Abstract

Embodiments disclosed herein relate to device testing using scan chains including various flip-flop devices in a multi-bit flip-flop configuration. A circuit device included herein includes a first flip-flop sub-circuit and a second flip-flop sub-circuit. The first flip-flop sub-circuit is coupled to receive a clock signal and an input, and the second flip-flop circuit is coupled to the first flip-flop sub-circuit. The first flip-flop sub-circuit includes an input sub-circuit, a first latch sub-circuit, a first latch tristate, a second latch sub-circuit, and a first output inverter. The second latch sub-circuit includes a first transmission gate, a first inverter, and a second inverter. The second flip-flop sub-circuit includes a second transmission gate, a first clock tristate, a third latch sub-circuit, a second latch tristate, a fourth latch sub-circuit, and a second output inverter. The fourth latch sub-circuit includes a third transmission gate, a third inverter, and a fourth inverter.

Inventors: Khawas; Arnab (Bangalore, IN), Pissay; Madhavan (Hyderabad, IN), Subbannavar; Badarish (Bangalore, IN)

Applicant: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

Family ID: 1000008820166

Assignee: Texas Instruments Incorporated (Dallas, TX)

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Primary Examiner: O Neill; Patrick

Background/Summary

TECHNICAL FIELD

(1) This relates generally to multi-bit flip-flop circuits, and more particularly, to performing scan chain testing therewith.

BACKGROUND

(2) In digital logic circuits, flip-flops and latches can be included to store state information and ensure proper sequencing for logic in electronics. The output of a flip-flop may transition (e.g., switch state from “0” to “1” and “1” to “0”) at specific times determined based on the state of a clock signal. Outside of these times, the flip-flop retains its current state. In this way, a flip-flop stores state data, such as a “0” or “1” based on the state of the clock signal and the state of a data input.

(3) As digital system functionality increases in complexity, individual standard cell area optimization for digital circuits, such as flip-flops and latches, becomes increasingly critical. To reduce design area and power consumption by digital circuits, existing solutions may include multi-bit flip-flops in place of single-bit flip-flops. Multi-bit flip-flops may be used to store multiple bit signals using a single clock signal, and thus, they can save area where multiple single-bit flip-flops may be used.

(4) Problematically, however, various multi-bit flip-flops existing today, include significant numbers of transistors despite reducing design area relative to single-bit flip-flops. Therefore, existing architectures of multi-bit flip-flops may fail to realize power and area savings for increasingly complex digital systems.

SUMMARY

(5) Various embodiments disclosed herein relate to multi-bit flip-flop circuits, and more particularly, to architecture of flip-flop circuits of multi-bit flip-flop circuits. In an example embodiment, a circuit device is provided. The circuit device includes a first flip-flop sub-circuit coupled to receive a clock signal and an input and a second flip-flop sub-circuit coupled to receive the clock signal and coupled to the first flip-flop sub-circuit. The first flip-flop sub-circuit comprises an input sub-circuit coupled to receive the clock signal and the input, a first latch sub-

circuit coupled to receive the clock signal and coupled to the input sub-circuit, a first latch tristate coupled to receive the clock signal and coupled to the first latch sub-circuit, a second latch sub-circuit coupled to receive the clock signal and coupled to the first latch tristate, and a first output inverter coupled to the first latch tristate and to the second latch sub-circuit. The second latch sub-circuit includes a first transmission gate coupled to the first latch tristate, the first output inverter, and the second flip-flop sub-circuit, a first inverter coupled to the first transmission gate, and a second inverter coupled to the first inverter, the first latch tristate, and the first output inverter. The second flip-flop sub-circuit comprises a second transmission gate coupled to receive the clock signal and coupled to the first transmission gate of the second latch sub-circuit, a first clock tristate coupled to receive the clock signal and coupled to the second transmission gate, a third latch sub-circuit coupled to receive the clock signal and coupled to the clock tristate, a second latch tristate coupled to receive the clock signal and coupled to the third latch sub-circuit, a fourth latch sub-circuit coupled to receive the clock signal and coupled to the second latch tristate, and a second output inverter coupled to the second latch tristate and to the fourth latch sub-circuit. The fourth latch sub-circuit includes a third transmission gate coupled to the second latch tristate and the second output inverter, a third inverter coupled to the third transmission gate, and a fourth inverter coupled to the third inverter, the second latch tristate, and the second output inverter.

(6) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. It may be understood that this Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A and 1B (collectively FIG. 1) illustrate an example multi-bit flip-flop circuit in accordance with an embodiment.

(2) FIG. 2 illustrates an example operating environment for executing logic with a multi-bit flip-flop circuit in accordance with an embodiment.

(3) The drawings are not necessarily drawn to scale. In the drawings, like reference numerals designate corresponding parts throughout the several views. In some embodiments, components or operations may be separated into different blocks or may be combined into a single block.

DETAILED DESCRIPTION

(4) Embodiments of the present disclosure are described in specific contexts, such as in multi-bit flip-flop circuits, primary-secondary latch circuits, digital logic and inverter circuitry, and the like. Some embodiments may use other circuits, digital logic components, topologies, and applications that exhibit increased transistor efficiency and other characteristics based on decreasing the number of transistors in a system or circuit.

(5) Discussed herein are enhanced components, systems, and architectures related to multi-bit flip-flop circuits used in digital systems or circuits to execute multiple bit-level operations on logic signals. In an electronic system, a flip-flop circuit can store logical states of inputs to the flip-flop circuit, such as scan data during a testing phase or input data during real-time operations. Flip-flops and latches are examples of logic components within digital circuitry that can store state data (e.g., “0” or “1”) based on inputs to the flip-flops (e.g., an input data signal) at times dictated by a clock signal. Some example flip-flops include one or more inverters that can be triggered by a clock signal to store the state of the input data signal.

(6) Multi-bit flip-flops may be included in a system to store such logical states among several different data inputs. Such multi-bit flip-flops may be tested by scan chaining several bits, or individual flip-flop circuits, of the multi-bit flip-flop circuit together. Scan data inputs may be

provided to a first bit flip-flop circuit in the multi-bit flip-flop circuit and measured at an output node of the multi-bit flip-flop circuit to ensure accuracy, efficiency, and power consumption of the multi-bit flip-flop circuit. Problematically, however, multi-bit flip-flop circuits include significant numbers of transistors for both data storage and scan testing purposes, despite being more power and area conservative relative to multiple single-bit flip-flop circuits.

(7) As disclosed herein, examples of a multi-bit flip-flop circuit are described that include a reduced number of transistors compared to existing multi-bit flip-flop circuit topologies. The multi-bit flip-flop circuit may include an input sub-circuit, primary and secondary latch sub-circuits, and an output inverter, among other components. In some examples, the secondary latch sub-circuits of each flip-flop bit in the multi-bit flip-flop circuit includes two inverters and a transmission gate, which may utilize a reduced number of transistors among the multi-bit flip-flop circuit.

Additionally, each input sub-circuit from the second bit, or second stage, of the multi-bit flip-flop circuit until the last bit of an n-bit multi-bit flip-flop circuit may include a scan transmission gate for the scan data input as opposed to a scan input multiplexer, which may further reduce the number of transistors among the multi-bit flip-flop circuit. In this way, with respect to the input sub-circuits, the transistor count of the multi-bit flip-flop circuit may be reduced by $(n-1)*2$ transistors, where n is the number of bits or stages in the multi-bit flip-flop circuit. Advantageously, such a topology can reduce latency within the flip-flop circuit during scan testing, latency within an overall digital circuit that implements logic based on the data stored by the multi-bit flip-flop circuit, power consumption by the multi-bit flip-flop circuit, and design cost by avoiding large numbers of transistors.

(8) In an example embodiment, a circuit device is provided. The circuit device includes a first flip-flop sub-circuit coupled to receive a clock signal and an input and a second flip-flop sub-circuit coupled to receive the clock signal and coupled to the first flip-flop sub-circuit. The first flip-flop sub-circuit comprises an input sub-circuit coupled to receive the clock signal and the input, a first latch sub-circuit coupled to receive the clock signal and coupled to the input sub-circuit, a first latch tristate coupled to receive the clock signal and coupled to the first latch sub-circuit, a second latch sub-circuit coupled to receive the clock signal and coupled to the first latch tristate, and a first output inverter coupled to the first latch tristate and to the second latch sub-circuit. The second latch sub-circuit includes a first transmission gate coupled to the first latch tristate, the first output inverter, and the second flip-flop sub-circuit, a first inverter coupled to the first transmission gate, and a second inverter coupled to the first inverter, the first latch tristate, and the first output inverter. The second flip-flop sub-circuit comprises a second transmission gate coupled to receive the clock signal and coupled to the first transmission gate of the second latch sub-circuit, a first clock tristate coupled to receive the clock signal and coupled to the second transmission gate, a third latch sub-circuit coupled to receive the clock signal and coupled to the clock tristate, a second latch tristate coupled to receive the clock signal and coupled to the third latch sub-circuit, a fourth latch sub-circuit coupled to receive the clock signal and coupled to the second latch tristate, and a second output inverter coupled to the second latch tristate and to the fourth latch sub-circuit. The fourth latch sub-circuit includes a third transmission gate coupled to the second latch tristate and the second output inverter, a third inverter coupled to the third transmission gate, and a fourth inverter coupled to the third inverter, the second latch tristate, and the second output inverter.

(9) In another example embodiment, a circuit device is provided. The circuit device includes a first flip-flop sub-circuit coupled to receive a clock signal and an input, and a second flip-flop sub-circuit coupled to receive the clock signal and coupled to the first flip-flop sub-circuit. The first flip-flop sub-circuit comprises an input sub-circuit coupled to receive the clock signal and the input, a first latch sub-circuit coupled to receive the clock signal and coupled to the input sub-circuit, a first latch tristate coupled to receive the clock signal and coupled to the first latch sub-circuit, a second latch sub-circuit coupled to receive the clock signal and coupled to the first latch tristate, and a first output inverter coupled to the first latch tristate and to the second latch sub-

circuit. The second latch sub-circuit includes a first transmission gate coupled to the first latch tristate, the first output inverter, and the second flip-flop sub-circuit, a first inverter coupled to the first transmission gate, and a second inverter coupled to the first inverter, the first latch tristate, and the first output inverter.

(10) In yet another example embodiment, a circuit device is provided that includes a first flip-flop sub-circuit coupled to receive a clock signal and an input, and a second flip-flop sub-circuit coupled to receive the clock signal and coupled to the first flip-flop sub-circuit. The second flip-flop sub-circuit comprises a first transmission gate coupled to receive the clock signal and coupled to a first transmission gate of a first latch sub-circuit; a first transmission gate coupled to receive the clock signal and coupled to the first flip-flop sub-circuit, a first clock tristate coupled to receive the clock signal and coupled to the first transmission gate, a first latch sub-circuit coupled to receive the clock signal and coupled to the first clock tristate, a first latch tristate coupled to receive the clock signal and coupled to the first latch sub-circuit, a second latch sub-circuit coupled to receive the clock signal and coupled to the first latch tristate, and a first output inverter coupled to the first latch tristate and to the second latch sub-circuit. The second latch sub-circuit includes a second transmission gate coupled to the first latch tristate and the first output inverter, a first inverter coupled to the second transmission gate, and a second inverter coupled to the first inverter, the first latch tristate, and the first output inverter.

(11) FIGS. 1A and 1B illustrate an example multi-bit flip-flop circuit in accordance with an embodiment. FIG. 1 includes multi-bit flip-flop circuit **100**, which includes flip-flop circuit **110-1**, flip-flop circuit **110-2**, and clock buffer **111**. Flip-flop circuit **110-1** further includes input sub-circuit **120**, primary latch sub-circuit **131**, latch clock tristate **140**, secondary latch sub-circuit **145**, and output inverter **155**. Flip-flop circuit **110-2** further includes input sub-circuit **160**, primary latch sub-circuit **171**, latch clock tristate **180**, secondary latch sub-circuit **185**, and output inverter **195**.

(12) Multi-bit flip-flop circuit **100** may be representative of a two-bit flip-flop circuit capable of resolving multiple bit-level signals (input functional data **107** and input functional data **108**) provided by a digital circuit, for example, based on clock signal **101** provided by clock buffer **111**. Each bit of multi-bit flip-flop circuit **100**, or flip-flop circuit **110-1** and flip-flop circuit **110-2**, may be used to store a state (e.g., “0”, “1”) of an input signal and provide the stored state data downstream to subsequent bits of multi-bit flip-flop circuit **100**. Accordingly, multi-bit flip-flop circuit **100** may be used to provide power improvements and area reduction, among other benefits, of a digital system.

(13) Multi-bit flip-flop circuit **100** includes flip-flop circuit **110-1**, flip-flop circuit **110-2**, and clock buffer **111**. Flip-flop circuit **110-1** may include a set of inputs coupled to receive inverted clock signal **102** and clock signal **103** from clock buffer **111**, scan data **104**, scan enable signal **105**, and inverted scan enable signal **106**, and input functional data **107**. Flip-flop circuit **110-1** may include an output to provide output **109-1** to flip-flop circuit **110-2**. Flip-flop circuit **110-2** may be coupled to receive inverted clock signal **102** and clock signal **103** from clock buffer **111**, scan data **104**, scan enable signal **105**, and inverted scan enable signal **106**, input functional data **108**, and output **109-1** from flip-flop circuit **110-1**. Flip-flop circuit **110-2** may be coupled to provide output **109-2** to a downstream circuit or system, such as another flip-flop circuit of multi-bit flip-flop circuit **100**. Clock buffer **111** may be coupled to receive clock signal **101** (e.g., from a timing or clock circuit) and output inverted clock signal **102** and clock signal **103**.

(14) Each of the input and output signals, such as clock signal **101**, inverted clock signal **102**, clock signal **103**, scan data **104**, scan enable signal **105**, inverted scan enable signal **106**, input functional data **107**, input functional data **108**, output **109-1**, and output data **109-2** may be electronic logic signals each including a value indicative of a logical low state (i.e., “0”) or a logical high state (i.e., “1”). For example, when clock signal **101** is high, inverted clock signal **102** may be low and clock signal **103** may be high. Similarly, when scan enable signal **105** is high, inverted scan enable signal **106** may be low. The state of scan enable signal **105** may be controlled by a controller or another

circuit, for example. When scan enable signal **105** is high, multi-bit flip-flop circuit **100** may enter a scan mode whereby components, and nodes thereof, of flip-flop circuits **110-1** and **110-2** may be tested for efficiency and accuracy. When scan enable signal **105** is low, multi-bit flip-flop circuit **100** may enter a data mode whereby multi-bit flip-flop circuit **100** may store state information of input functional data **107** and input functional data **108** for use by systems and digital circuits coupled to multi-bit flip-flop circuit **100**.

(15) The logical states of signals within multi-bit flip-flop circuit **100** and operations of multi-bit flip-flop circuit **100** may be based on the logical state of clock signal **103** provided to nodes of multi-bit flip-flop circuit **100** by clock buffer **111**. Clock buffer **111** includes transistors **112**, **113**, **114**, and **115**. Transistors **112** and **114** may be p-channel (or p-type) metal-oxide semiconductor field-effect transistors (MOSFETs) (also referred to as PMOS), and transistors **113** and **115** may be n-channel (or n-type) MOSFETs (also referred to as NMOS). Each of transistors **112**, **113**, **114**, and **115** may include a drain a source, and a gate. The gates of transistors **112** and **113** may be coupled together and coupled to receive clock signal **101** from a timing or clock circuit. The drains of transistors **112** and **113** may also be coupled together and coupled to the gates of transistors **114** and **115**, which may be coupled to receive inverted clock signal **102** from transistors **112** and **113**. The sources of transistors **112** and **114** may be coupled to receive a power from an internal power supply (e.g., V.sub.dd), and the sources of transistors **113** and **115** may be coupled to individual ground nodes. The drains of transistors **114** and **115** may also be coupled together and may be coupled to provide clock signal **103** to nodes of flip-flop circuits **110-1** and **110-2**. In effect, transistors **112** and **113** may form a first inverter of clock buffer **111**, and transistors **114** and **115** may form a second inverter of clock buffer **111**. Accordingly, inverted clock signal **102** and clock signal **103** can be produced by clock buffer **111** and provided to flip-flop circuits **110-1** and **110-2**.

(16) Flip-flop circuit **110-1** includes various components coupled to receive signals from clock buffer **111** and coupled to provide outputs to flip-flop circuit **110-2**. Specifically, flip-flop circuit **110-1** includes input sub-circuit **120**, primary latch sub-circuit **131**, latch clock tristate **140**, secondary latch sub-circuit **145**, and output inverter **155**. Input sub-circuit **120** may include a set of inputs coupled to receive inverted clock signal **102** and clock signal **103** from clock buffer **111**, scan data **104**, scan enable signal **105**, and inverted scan enable signal **106** from a controller or another circuit, and input functional data **107** from another circuit. Input sub-circuit **120** may include an output defined by a node between transistor **127** and transistor **128** described below. Primary latch sub-circuit **131** may include an input coupled to the output of input sub-circuit **120** and a set of inputs coupled to clock buffer **111**. Primary latch sub-circuit **131** may include an output defined by a node between transistors **138** and **139** and coupled to the gates of transistors **133** and **136** described below. Latch clock tristate **140** may include a set of inputs coupled to the output of primary latch sub-circuit **131** and a set of inputs coupled to clock buffer **111**. . . . Latch clock tristate **140** may include an output defined by a node between transistors **142** and **143** described below. Secondary latch sub-circuit **145** may include a set of inputs coupled to the output of latch clock tristate **140** and a set of inputs coupled to clock buffer **111**. Secondary latch sub-circuit **145** may include an output defined by a node between transistors **150** and **151** described below. A set of the inputs of secondary latch sub-circuit **145** may also be coupled to an input of output inverter **155**. Output inverter **155** may include an input coupled to the output of latch clock tristate **140** and may include an output defined by a node between transistors **156** and **157** described below.

(17) Input sub-circuit **120** may include transistors **121**, **122**, **123**, **124**, **125**, **126**, **127**, **128**, **129**, and **130**, which, together, may form an input scan multiplexer (MUX) and an input clock tristate. Each of the transistors of input sub-circuit **120** may be an NMOS or a PMOS, and each may include a gate, a drain, and a source. In various examples, transistors **121**, **122**, **123**, and **124** may form the input scan MUX, and transistors **125**, **126**, **127**, **128**, **129**, and **130** may form the input clock tristate. The gates of each of these transistors may be coupled to receive an input signal. The gates of transistors **121** and **124** may be coupled to receive scan data **104**, the gates of transistors **122** and

130 may be coupled to receive inverted scan enable signal **106**, the gates of transistors **123** and **125** may be coupled to receive scan enable signal **105**, the gate of transistor **128** may be coupled to receive inverted clock signal **102**, the gate of transistor **127** may be coupled to receive clock signal **103**, and the gates of transistors **126** and **129** may be coupled to receive input functional data **107**. The source of transistors **121** may be coupled to receive a power from a power supply. The drain of transistor **121** may be coupled to the source of transistor **122**, and the drain of transistor **122** may be coupled to the drain of transistor **126** and the source of transistor **127**. The source of transistor **124** may be coupled to a ground node, the drain of transistor **124** may be coupled to the source of transistor **123**, and the drain of transistor **123** may be coupled to the source of transistor **128** and the drain of transistor **129**. The source of transistor **125** may also be coupled to receive a power from the power supply, and the drain of transistor **125** may be coupled to the source of transistor **126**. The drain of transistor **126** may be coupled to the source of transistor **127**. The drain of transistor **127** may be coupled to the drain of transistor **128**, both of which may be further coupled to primary latch sub-circuit **131**. The source of transistor **128** may be coupled to the drain of transistor **129**. The source of transistor **129** may be coupled to the drain of transistor **130**. The source of transistor **130** may be coupled to another ground node.

(18) Primary latch sub-circuit **131** is representative of a first or primary control flip-flop (e.g., a master latch) that can control operations of secondary latch sub-circuit **145**, or a second control flip-flop following the first control flip-flop (e.g., a slave latch). Primary latch sub-circuit **131** can store a state value (e.g., “0” or “1”) of input functional data **107** or scan data **104** (based on the state of scan enable signal **105**) and provide the state value to latch clock tristate **140** for further distribution to secondary latch sub-circuit **145** and output inverter **155**. Primary latch sub-circuit **131** can be configured to store a subsequent state value of input functional data **107** or scan data **104** when the state of input functional data **107** or scan data **104** change to a different state relative to its stored state. Following the latching of a new state of the data input, primary latch sub-circuit **131** can provide the new state value to secondary latch sub-circuit **145** via latch clock tristate **140** for storage.

(19) In this example, primary latch sub-circuit **131** includes feedback tristate **132** and inverter **137**, which each include a number of NMOS and PMOS transistors. Feedback tristate **132** includes transistors **133**, **134**, **135**, and **136**, and inverter **137** includes transistors **138** and **139**. Both feedback tristate **132** and inverter **137** are coupled to receive a signal from input sub-circuit **120**. Transistor **133** includes a source coupled to receive power from a power supply, a gate coupled to a gate of transistor **136**, and a drain coupled to a source of transistor **134**. Transistor **134** includes a gate coupled to receive inverted clock signal **102** and a drain coupled to a drain of transistor **135**, both of which are further coupled to the drains of transistors **127** and **128**. Transistor **125** also includes a gate coupled to receive clock signal **103** and a source coupled to a drain of transistor **136**, which also includes a gate coupled to the gate of transistor **133** and a source coupled to a ground node. The gates of transistors **133** and **136** are coupled to latch clock tristate **140** at gates of transistors **141** and **144** of latch clock tristate **140**. Transistors **138** and **139** each include gates coupled to the drains of transistors **127** and **128** and to the drains of transistors **134** and **135**. Transistor **138** also includes a source coupled to receive power from the power supply and a drain coupled to a drain of transistor **139**. The source of transistor **139** is coupled to a ground node. The drains of transistors **138** and **139** are coupled to the gates of transistors **133** and **136**, and thus, are also coupled to latch clock tristate **140**.

(20) Latch clock tristate **140** includes transistors **141**, **142**, **143**, and **144**. Transistors **141** and **142** may be PMOS transistors, while transistors **143** and **144** may be NMOS transistors. Transistor **141** may include a source coupled to receive power from a power supply, a gate coupled to a gate of transistor **144** and further coupled to primary latch sub-circuit **131**, and a drain coupled to a source of transistor **142**. Transistor **142** may also include a gate coupled to receive inverted clock signal **102** from clock buffer **111** and a drain coupled to a drain of transistor **143**, both of which may be

coupled to secondary latch sub-circuit **145** and output inverter **155**. Transistor **143** may include a gate coupled to receive clock signal **103** from clock buffer **111** and a source coupled to a drain of transistor **144**. The source of transistor **144** may be coupled to a ground node.

(21) Secondary latch sub-circuit **145** includes inverter **146**, transmission gate **149**, and inverter **152**, each of which include a number of PMOS and NMOS transistors. Inverter **146** includes transistors **147** and **148**, transmission gate **149** includes transistors **150** and **151**, and inverter **152** includes transistors **153** and **154**. Transistor **147** includes a source coupled to receive a power from a power supply, a gate coupled to a gate of transistor **148**, and a drain coupled to a drain of transistor **148**. Transistor **148** also includes a source coupled to a ground node. The gates of transistors **147** and **148** are coupled to latch clock tristate **140**, transmission gate **149**, and to output inverter **155**. The drains of transistors **147** and **148** are coupled to gates of transistors **153** and **154** of inverter **152**. Transistor **150** includes a gate coupled to receive clock signal **103** from clock buffer **111**, a drain coupled to a drain of transistor **151**, and a source coupled to a source of transistor **151**. Transistor **151** includes a gate coupled to receive inverted clock signal **102** from clock buffer **111**. The drains of transistors **150** and **151** are further coupled to latch clock tristate **140**, inverter **146**, and output inverter **155**. The sources are further coupled to input sub-circuit **160** of flip-flop circuit **110-2**, or more particularly, to transistors **162** and **163** of scan transmission gate **161** of flip-flop circuit **110-2**, and to transistors **153** and **154** of inverter **152**. In some examples, the drains of transistors **150** and **151** may instead be coupled to input sub-circuit **160** of flip-flop circuit **110-2**. Transistor **153** includes a source coupled to receive power from a power supply, a gate coupled to a gate of transistor **154**, and a drain coupled to a drain of transistor **154**. Transistor **154** also includes a source coupled to a ground node. The gates of transistors **153** and **154** are coupled to transistors **147** and **148** of inverter **146**.

(22) Output inverter **155** includes transistors **156** and **157** coupled to receive signals from latch clock tristate **140** and secondary latch sub-circuit **145** and coupled to provide output **109-1** downstream. Transistor **156** includes a gate coupled to a gate of transistor **157**, a source coupled to receive power from a power supply, and a drain coupled to a drain of transistor **157**. Transistor **157** also includes a source coupled to a ground node. Output **109-1** can be provided to one or more downstream systems or circuits via the drains of transistors **156** and **157**.

(23) Flip-flop circuit **110-2** may be representative of a second bit flip-flop of multi-bit flip-flop circuit **100** that includes various components coupled to receive signals from clock buffer **111** and flip-flop circuit **110-2** and coupled to provide outputs to a further flip-flop sub-circuit. Flip-flop circuit **110-2** may include input sub-circuit **160**, primary latch sub-circuit **171**, latch clock tristate **180**, secondary latch sub-circuit **185**, and output inverter **195**. Input sub-circuit **160** may include a set of inputs coupled to receive inverted clock signal **102** and clock signal **103** from clock buffer **111**, scan data **104**, scan enable signal **105**, and inverted scan enable signal **106** from a controller or another circuit, and input functional data **108** from another circuit. Input sub-circuit **160** may include an output defined by a node between transistor **167** and transistor **168** described below. Primary latch sub-circuit **171** may include an input coupled to the output of input sub-circuit **160** and a set of inputs coupled to clock buffer **111**. Primary latch sub-circuit **171** may include an output defined by a node between transistors **178** and **179** and coupled to the gates of transistors **173** and **176** described below. Latch clock tristate **180** may include an input coupled to the output of primary latch sub-circuit **171** and a set of inputs coupled to clock buffer **111**. Latch clock tristate **180** may include an output defined by a node between transistors **182** and **183** described below. Secondary latch sub-circuit **185** may include a set of inputs coupled to the output of latch clock tristate **180** and a set of inputs coupled to clock buffer **111**. A set of the inputs of secondary latch sub-circuit **185** may also be coupled to output inverter **195**. Secondary latch sub-circuit **185** may include an output defined by a node between transistors **190** and **191** described below. Output inverter **195** may include an input coupled to the output of latch clock tristate **180** and may include an output defined by a node between transistors **196** and **197** described below.

(24) Input sub-circuit **160** may include scan transmission gate **161**, which includes transistors **162** and **163**, and input clock tristate **164**, which includes transistors **165**, **166**, **167**, **168**, **169**, and **170**. Each of the transistors of input sub-circuit **160** may be an NMOS or a PMOS, and each may include a gate, a drain, and a source. The gates of transistors **162** and **170** may be coupled to receive inverted scan enable signal **106**, the gates of transistors **163** and **165** may be coupled to receive scan enable signal **105**, the gates of transistors **166** and **169** may be coupled to receive input functional data **108**, the gate of transistor **167** may be coupled to receive clock signal **103**, and the gate of transistor **168** may be coupled to receive inverted clock signal **102**. The source of transistor **162** may be coupled to the drain of transistor **166** and to the source of transistor **167**. The drain of transistor **162** may be coupled to the drain of transistor **163**, both of which may be coupled to receive a signal having the same state as output **109-1** from transmission gate **149** of flip-flop circuit **110-1**. The source of transistor **163** may be coupled to the source of transistor **168** and to the drain of transistor **169**. The source of transistor **165** may be coupled to receive a power from a power supply, and the drain of transistor **165** may be coupled to the source of transistor **166**. The drain of transistor **166** may be coupled to the source of transistor **167**. The drain of transistor **167** may be coupled to the drain of transistor **168**. The source of transistor **168** may be coupled to the drain of transistor **169**. The source of transistor **169** may be coupled to the transistor drain of transistor **170**. The source of transistor **170** may be coupled to a ground node. Input sub-circuit **160** may be coupled to primary latch sub-circuit **171** via the drains of transistors **167** and **168**.

(25) Primary latch sub-circuit **171** is representative of a first or primary control flip-flop (e.g., a master latch) that can control operations of secondary latch sub-circuit **185**, or a second control flip-flop following the first control flip-flop (e.g., a slave latch). Primary latch sub-circuit **171** can store a state value (e.g., “0” or “1”) of input functional data **108** or scan data **104** (based on the state of scan enable signal **105**) and provide the state value to latch clock tristate **180** for further distribution to secondary latch sub-circuit **185** and output inverter **195**. Primary latch sub-circuit **171** can be configured to store a subsequent state value of input functional data **108** or scan data **104** when the state of input functional data **108** or scan data **104** change to a different state relative to its stored state. Following the latching of a new state of the data input, primary latch sub-circuit **171** can provide the new state value to secondary latch sub-circuit **185** via latch clock tristate **180** for storage.

(26) In this example, primary latch sub-circuit **171** includes feedback tristate **172** and inverter **177**, which each include a number of NMOS and PMOS transistors. Feedback tristate **172** includes transistors **173**, **174**, **175**, and **176**, and inverter **177** includes transistors **178** and **179**. Both feedback tristate **172** and inverter **177** are coupled to receive a signal from input sub-circuit **120**. Transistor **173** includes a source coupled to receive power from a power supply, a gate coupled to a gate of transistor **176**, and a drain coupled to a source of transistor **174**. Transistor **174** includes a gate coupled to receive inverted clock signal **102** and a drain coupled to a drain of transistor **175**, both of which are further coupled to the drains of transistors **127** and **128**. Transistor **125** also includes a gate coupled to receive clock signal **103** and a source coupled to a drain of transistor **176**, which also includes a gate coupled to the gate of transistor **173** and a source coupled to a ground node. The gates of transistors **173** and **176** are coupled to latch clock tristate **180** at gates of transistors **181** and **184** of latch clock tristate **180**. Transistors **178** and **179** each include gates coupled to the drains of transistors **167** and **168** and to the source of transistor **174** and to the drain of transistor **175**. Transistor **178** also includes a source coupled to receive power from the power supply and a drain coupled to a drain of transistor **179**. The source of transistor **179** is coupled to a ground node. The drains of transistors **178** and **179** are coupled to the gates of transistors **173** and **176**, and thus, are also coupled to latch clock tristate **180**.

(27) Latch clock tristate **180** includes transistors **181**, **182**, **183**, and **184**. Transistors **181** and **182** may be PMOS transistors, while transistors **183** and **184** may be NMOS transistors. Transistor **181** may include a source coupled to receive power from a power supply, a gate coupled to a gate of

transistor **184** and further coupled to primary latch sub-circuit **171**, and a drain coupled to a source of transistor **182**. Transistor **182** may also include a gate coupled to receive inverted clock signal **102** from clock buffer **111** and a drain coupled to a drain of transistor **183**, both of which may be coupled to secondary latch sub-circuit **185** and output inverter **195**. Transistor **183** may include a gate coupled to receive clock signal **103** from clock buffer **111** and a source coupled to a drain of transistor **184**. The source of transistor **184** may be coupled to a ground node.

(28) Secondary latch sub-circuit **185** includes inverter **186**, transmission gate **189**, and inverter **192**, each of which include a number of PMOS and NMOS transistors. Inverter **186** includes transistors **187** and **188**, transmission gate **189** includes transistors **190** and **191**, and inverter **192** includes transistors **193** and **194**. Transistor **187** includes a source coupled to receive a power from a power supply, a gate coupled to a gate of transistor **188**, and a drain coupled to a drain of transistor **188**. Transistor **188** also includes a source coupled to a ground node. The gates of transistors **187** and **188** are coupled to latch clock tristate **180**, transmission gate **189**, and to output inverter **195**. The drains of transistors **187** and **188** are coupled to gates of transistors **193** and **194** of inverter **192**. Transistor **190** includes a gate coupled to receive clock signal **103** from clock buffer **111**, a drain coupled to a drain of transistor **191**, and a source coupled to a source of transistor **191**. Transistor **191** includes a gate coupled to receive inverted clock signal **102** from clock buffer **111**. The drains of transistors **190** and **191** are further coupled to latch clock tristate **180**, inverter **186**, and output inverter **195**. The sources may be further coupled to a further flip-flop circuit (e.g., to another flip-flop bit of multi-bit flip-flop circuit **100**) and to transistors **193** and **194** of inverter **192**. In some examples, the drains of transistors **190** and **191** may instead be coupled to the further flip-flop circuit. Transistor **193** includes a source coupled to receive power from a power supply, a gate coupled to a gate of transistor **194**, and a drain coupled to a drain of transistor **194**. Transistor **194** also includes a source coupled to a ground node. The gates of transistors **193** and **194** are coupled to transistors **187** and **188** of inverter **186**.

(29) Output inverter **195** includes transistors **196** and **197** coupled to receive signals from latch clock tristate **180** and secondary latch sub-circuit **185** and coupled to provide output **109-2** downstream. Transistor **196** includes a gate coupled to a gate of transistor **197**, a source coupled to receive power from a power supply, and a drain coupled to a drain of transistor **197**. Transistor **197** also includes a source coupled to a ground node. Output **109-2** can be provided to one or more downstream systems or circuits via the drains of transistors **196** and **197**.

(30) In various examples, the topology of multi-bit flip-flop circuit **100** illustrated and described with respect to FIG. **1** may reduce overall power consumption as a single clock buffer (clock buffer **111**) may be used for both flip-flop circuits **110-1** and **110-2**, cell-level area as multiple flip-flop bits can be stitched together using a shared clock buffer and scan inverter, and design area as reduced numbers of transistors may be used, especially in the secondary latch sub-circuits and in the input sub-circuits, relative to existing multi-bit flip-flop circuit designs.

(31) In some examples, other types of transistors may be used instead or in addition to the NMOS and PMOS transistors shown and described herein. In an example, the primary latch sub-circuits and secondary latch sub-circuits may have different topologies individually and/or with respect to each other. For example, each latch sub-circuit may include a different number of inverters, different types of transistors, or the like.

(32) FIG. **2** illustrates an example operating environment for executing logic with a multi-bit flip-flop circuit in accordance with an embodiment. FIG. **2** shows operating environment **200**, which includes timing circuit **205**, logic circuit **210**, multi-bit flip-flop circuit **100**, which includes flip-flop circuit **110-1** and flip-flop circuit **110-2**, and logic circuit **225**.

(33) Logic circuit **210** is representative of a digital circuit, digital logic device, or combination or variation of electronic and logic elements capable of performing logic steps in accordance with a clock signal, such as one produced by timing circuit **205** (e.g., clock signal **101** of FIG. **1**). Logic circuit **210** can produce a data signal, a test signal, a scan signal, or another signal (e.g., scan data

104, scan enable signal **105**, input functional data **107**, input functional data **108**) and provide one or more of the signals to flip-flop circuits **110-1** and **110-2** of multi-bit flip-flop circuit **100**. For example, logic circuit **210** can provide input functional data **107** and scan data **104** to flip-flop circuit **110-1** and input functional data **108** to flip-flop circuit **110-2**.

(34) Multi-bit flip-flop circuit **100** may be representative of a two-bit flip-flop circuit capable of storing multiple bit-level signals (input functional data **107** and input functional data **108**) provided by logic circuit **210** based on a clock signal provided by timing circuit **205** (via a clock signal and/or a scan clock signal) and outputting multiple bit-level signals (outputs **109-1** and **109-2**). Each bit of multi-bit flip-flop circuit **100**, or flip-flop circuit **110-1** and flip-flop circuit **110-2**, may be used to store a state (e.g., “0”, “1”) of an input signal and provide the stored state data downstream to subsequent bits of multi-bit flip-flop circuit **100** and to other digital circuits, such as logic circuit **225**. Accordingly, multi-bit flip-flop circuit **100** may be used to provide power improvements and area reduction, among other benefits, of a digital system.

(35) Flip-flop circuits **110-1** and **110-2** may be representative of bit flip-flops of a multi-bit flip-flop circuit **100** capable of resolving and storing logical states of signals. Each flip-flop circuit of multi-bit flip-flop circuit **100** may include a data path and a scan path for resolving data or testing operations of the circuits, respectively. For example, when storing data, flip-flop circuit **110-1** may be coupled to receive input functional data **107** from logic circuit **210** and provide output **109-1** to logic circuit **225**, and flip-flop circuit **110-2** may be coupled to receive input functional data **108** from logic circuit **210** and provide output **109-2** to logic circuit **225**. When performing a scan of multi-bit flip-flop circuit **100**, flip-flop circuit **110-1** may be coupled to receive scan data **104** from logic circuit **210** and provide scan data output **201** to flip-flop circuit **110-2**. Flip-flop circuit **110-2** can then provide scan data output **202** to logic circuit **225**. Flip-flop circuits **110-1** and **110-2** may also be configured to receive a clock signal from timing circuit **205** (e.g., a clock buffer of multi-bit flip-flop circuit **100** (e.g., clock buffer **111**)). The flip-flop circuits can store values of the electronic signals and provide outputs indicative of the values to subsequent flip-flop circuits and/or to logic circuit **225**. In various examples, flip-flop circuits **110-1** and **110-2** may employ various transistors and topologies, such as those described above, to perform such functions.

(36) Logic circuit **225** is also representative of a digital circuit, digital logic device, or combination or variation of electronic and logic elements capable of performing logic steps in accordance with a clock signal, such as the clock signal produced by timing circuit **205**.

(37) While some examples provided herein are described in the context of a multi-bit flip-flop circuit, sub-circuit, system, subsystem, component, device, architecture, or environment, it should be understood that the gates, latches, flip-flops, logic elements, and other circuits, systems, and methods described herein are not limited to such embodiments and may apply to a variety of other processes, systems, applications, devices, and the like, such as other circuits, logic devices, latches, transistors, and the like, in the context of increasing transistor resolution efficiency, among other benefits, for example. Accordingly, aspects of the present invention may be embodied as other systems, methods, and other configurable systems.

(38) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” or any variant thereof means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or,” in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of

the items in the list, and any combination of the items in the list.

(39) The phrases “in some embodiments,” “according to some embodiments,” “in the embodiments shown,” “in other embodiments,” and the like generally mean the particular feature, structure, or characteristic following the phrase is included in at least one implementation of the present technology, and may be included in more than one implementation. In addition, such phrases do not necessarily refer to the same embodiments or different embodiments.

(40) The above Detailed Description of examples of the technology is not intended to be exhaustive or to limit the technology to the precise form disclosed above. While specific examples for the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative implementations may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or subcombinations. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed or implemented in parallel or may be performed at different times. Further any specific numbers noted herein are only examples: alternative implementations may employ differing values or ranges.

(41) The teachings of the technology provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various examples described above can be combined to provide further implementations of the technology. Some alternative implementations of the technology may include not only additional elements to those implementations noted above, but also may include fewer elements.

(42) These and other changes can be made to the technology in light of the above Detailed Description. While the above description describes certain examples of the technology, and describes the best mode contemplated, no matter how detailed the above appears in text, the technology can be practiced in many ways. Details of the system may vary considerably in its specific implementation, while still being encompassed by the technology disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the technology should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the technology with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the technology to the specific examples disclosed in the specification, unless the above Detailed Description section explicitly defines such terms. Accordingly, the actual scope of the technology encompasses not only the disclosed examples, but also all equivalent ways of practicing or implementing the technology under the claims.

(43) To reduce the number of claims, certain aspects of the technology are presented below in certain claim forms, but the applicant contemplates the various aspects of the technology in any number of claim forms. For example, while only one aspect of the technology is recited as a computer-readable medium claim, other aspects may likewise be embodied as a computer-readable medium claim, or in other forms, such as being embodied in a means-plus-function claim. Any claims intended to be treated under 35 U.S.C. § 112 (f) will begin with the words “means for” but use of the term “for” in any other context is not intended to invoke treatment under 35 U.S.C. § 112 (f). Accordingly, the applicant reserves the right to pursue additional claims after filing this application to pursue such additional claim forms, in either this application or in a continuing application.

Claims

1. A circuit device, comprising: a first flip-flop sub-circuit coupled to receive a clock signal and an input; and a second flip-flop sub-circuit coupled to receive the clock signal and coupled to the first flip-flop sub-circuit; wherein the first flip-flop sub-circuit comprises: an input sub-circuit coupled to receive the clock signal and the input; a first latch sub-circuit coupled to receive the clock signal and coupled to the input sub-circuit; a first latch tristate coupled to receive the clock signal and coupled to the first latch sub-circuit; a second latch sub-circuit coupled to receive the clock signal and coupled to the first latch tristate; and a first output inverter coupled to the first latch tristate and to the second latch sub-circuit; wherein the second latch sub-circuit includes: a first transmission gate coupled to the first latch tristate, the first output inverter, and the second flip-flop sub-circuit; a first inverter coupled to the first transmission gate; and a second inverter coupled to the first inverter, the first latch tristate, and the first output inverter; and wherein the second flip-flop sub-circuit comprises: a second transmission gate coupled to the first transmission gate of the second latch sub-circuit; a first clock tristate coupled to receive the clock signal and coupled to the second transmission gate; a third latch sub-circuit coupled to receive the clock signal and coupled to the first clock tristate; a second latch tristate coupled to receive the clock signal and coupled to the third latch sub-circuit; a fourth latch sub-circuit coupled to receive the clock signal and coupled to the second latch tristate; and a second output inverter coupled to the second latch tristate and to the fourth latch sub-circuit; wherein the fourth latch sub-circuit includes: a third transmission gate coupled to the second latch tristate and the second output inverter; a third inverter coupled to the third transmission gate; and a fourth inverter coupled to the third inverter, the second latch tristate, and the second output inverter.
2. The circuit device of claim 1, wherein the first flip-flop sub-circuit is a first bit flip-flop in a multi-bit flip-flop circuit, and wherein the second flip-flop sub-circuit is a second bit flip-flop in the multi-bit flip-flop circuit.
3. The circuit device of claim 2, wherein the multi-bit flip-flop circuit comprises a plurality of bits.
4. The circuit device of claim 1, wherein the input sub-circuit of the first flip-flop sub-circuit comprises an input scan multiplexer and a second clock tristate, wherein the input scan multiplexer is coupled to receive the input, and wherein the second clock tristate is coupled to the input scan multiplexer and coupled to receive the clock signal.
5. The circuit device of claim 1, wherein the first latch sub-circuit comprises a first feedback tristate and a fifth inverter, wherein the first feedback tristate is coupled to receive the clock signal and coupled to the input sub-circuit, and wherein the fifth inverter is coupled to the input sub-circuit.
6. The circuit device of claim 1, wherein the third latch sub-circuit comprises a second feedback tristate and a sixth inverter, wherein the second feedback tristate is coupled to receive the clock signal and coupled to the first clock tristate, and wherein the sixth inverter is coupled to the first clock tristate.
7. The circuit device of claim 1, wherein the first flip-flop sub-circuit and the second flip sub-circuit comprise a plurality of n-type transistors and a plurality of p-type transistors.
8. A circuit device, comprising: a first flip-flop sub-circuit coupled to receive a clock signal and an input; and a second flip-flop sub-circuit coupled to receive the clock signal and coupled to the first flip-flop sub-circuit; wherein the first flip-flop sub-circuit comprises: an input sub-circuit coupled to receive the clock signal and the input; a first latch sub-circuit coupled to receive the clock signal and coupled to the input sub-circuit; a first latch tristate coupled to receive the clock signal and coupled to the first latch sub-circuit; a second latch sub-circuit coupled to receive the clock signal and coupled to the first latch tristate; and a first output inverter coupled to the first latch tristate and to the second latch sub-circuit; wherein the second latch sub-circuit includes: a first transmission gate coupled to the first latch tristate, the first output inverter, and the second flip-flop sub-circuit; a first inverter coupled to the first transmission gate; and a second inverter coupled to the first

inverter, the first latch tristate, and the first output inverter.

9. The circuit device of claim 8, wherein the second flip-flop sub-circuit comprises: a second transmission gate coupled to the first transmission gate of the second latch sub-circuit; a first clock tristate coupled to receive the clock signal and coupled to the second transmission gate; a third latch sub-circuit coupled to receive the clock signal and coupled to the second transmission gate; a second latch tristate coupled to receive the clock signal and coupled to the third latch sub-circuit; a fourth latch sub-circuit coupled to receive the clock signal and coupled to the second latch tristate; and a second output inverter coupled to the second latch tristate and to the fourth latch sub-circuit; wherein the fourth latch sub-circuit includes: a third transmission gate coupled to the second latch tristate and the second output inverter; a third inverter coupled to the third transmission gate; and a fourth inverter coupled to the third inverter, the second latch tristate, and the second output inverter.

10. The circuit device of claim 9, wherein the first flip-flop sub-circuit is a first bit flip-flop in a multi-bit flip-flop circuit, and wherein the second flip-flop sub-circuit is a second bit flip-flop in the multi-bit flip-flop circuit.

11. The circuit device of claim 10, wherein the multi-bit flip-flop circuit comprises a plurality of bits.

12. The circuit device of claim 9, wherein the third latch sub-circuit comprises a second feedback tristate and a sixth inverter, wherein the second feedback tristate is coupled to receive the clock signal and coupled to the first clock tristate, and wherein the sixth inverter is coupled to the first clock tristate.

13. The circuit device of claim 9, wherein the first flip-flop sub-circuit and the second flip sub-circuit comprise a plurality of n-type transistors and a plurality of p-type transistors.

14. The circuit device of claim 8, wherein the input sub-circuit of the first flip-flop sub-circuit comprises an input scan multiplexer and a clock tristate, wherein the input scan multiplexer is coupled to receive the input, and wherein the clock tristate is coupled to the input scan multiplexer and coupled to receive the clock signal.

15. The circuit device of claim 8, wherein the first latch sub-circuit comprises a first feedback tristate and a fifth inverter, wherein the first feedback tristate is coupled to receive the clock signal and coupled to the input sub-circuit, and wherein the fifth inverter is coupled to the input sub-circuit.

16. A circuit device, comprising: a first flip-flop sub-circuit coupled to receive a clock signal and an input; and a second flip-flop sub-circuit coupled to receive the clock signal and coupled to the first flip-flop sub-circuit; wherein the second flip-flop sub-circuit comprises: a first transmission gate coupled to the first flip-flop sub-circuit; a first clock tristate coupled to receive the clock signal and coupled to the first transmission gate; a first latch sub-circuit coupled to receive the clock signal and coupled to the first clock tristate; a first latch tristate coupled to receive the clock signal and coupled to the first latch sub-circuit; a second latch sub-circuit coupled to receive the clock signal and coupled to the first latch tristate; and a first output inverter coupled to the first latch tristate and to the second latch sub-circuit; wherein the second latch sub-circuit includes: a second transmission gate coupled to the first latch tristate and the first output inverter; a first inverter coupled to the second transmission gate; and a second inverter coupled to the first inverter, the first latch tristate, and the first output inverter.

17. The circuit device of claim 16, wherein the first flip-flop sub-circuit comprises: an input sub-circuit coupled to receive the clock signal and the input; a third latch sub-circuit coupled to receive the clock signal and coupled to the input sub-circuit; a second latch tristate coupled to receive the clock signal and coupled to the third latch sub-circuit; a fourth latch sub-circuit coupled to receive the clock signal and coupled to the second latch tristate; and a second output inverter coupled to the second latch tristate and to the fourth latch sub-circuit; wherein the fourth latch sub-circuit includes: a third transmission gate coupled to the second latch tristate, the second output inverter, and the second flip-flop sub-circuit; a third inverter coupled to the third transmission gate; and a

- fourth inverter coupled to the third inverter, the second latch tristate, and the second output inverter.
18. The circuit device of claim 17, wherein the first flip-flop sub-circuit is a first bit flip-flop in a multi-bit flip-flop circuit, and wherein the second flip-flop sub-circuit is a second bit flip-flop in the multi-bit flip-flop circuit.
19. The circuit device of claim 17, wherein the input sub-circuit of the first flip-flop sub-circuit comprises an input scan multiplexer and a second clock tristate, wherein the input scan multiplexer is coupled to receive the input, and wherein the second clock tristate is coupled to the input scan multiplexer and coupled to receive the clock signal.
20. The circuit device of claim 19, wherein: the first latch sub-circuit comprises a first feedback tristate coupled to receive the clock signal and coupled to the input sub-circuit and a fifth inverter coupled to the input sub-circuit; and the third latch sub-circuit comprises a second feedback tristate coupled to receive the clock signal and coupled to the second clock tristate and a sixth inverter coupled to the second clock tristate.
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